

From Policy to Proof

How Aurora-Lens Can Strengthen Runtime Evidence within an ISO/IEC 42001 Artificial Intelligence Management System

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Abstract

ISO/IEC 42001:2023 establishes requirements for an artificial intelligence management system (AIMS): the organisational framework through which AI policy, responsibility, risk, impact, operation, monitoring, audit and improvement are governed. It is intentionally technology-neutral. It does not prescribe a particular control between model generation and consequential use, nor a technical architecture for proving that a control operated in an individual case.

That implementation freedom creates an evidentiary question. When an AI output contributes to a clinical record, credit decision, legal document or operational action, can the organisation show only that a control was designed, or can it also show what the control did at the relevant moment?

This paper examines Aurora-Lens as one possible runtime evidence layer within an AIMS. Aurora-Lens evaluates candidate model output before release, produces governed non-commitment outcomes where the basis for action is absent or defective, preserves governing state across interactions, and generates a tamper-evident decision record. These capabilities do not establish ISO/IEC 42001 conformity by themselves. They can, however, provide contemporaneous technical evidence that selected policies and risk treatments were executed in particular instances.

1. The question beneath conformity

ISO/IEC 42001:2023 specifies requirements for establishing, implementing, maintaining and continually improving an AIMS. ISO describes it as a management-system standard for organisations that develop, provide or use AI systems. Its scope is organisational: policies, objectives, roles, processes, risk management and continual improvement.

This paper addresses a narrower architectural question:

What evidence exists at the point where model output is permitted to become consequential?

An organisation may possess a carefully designed AIMS and still rely on procedures that are difficult to reconstruct at decision level. A policy may require human review. A risk treatment plan may require current authority. An impact assessment may identify a class of harm. An audit may later confirm that the relevant procedures existed.

Those facts matter. They do not, on their own, establish what occurred when one particular output was admitted, revised, contained or refused.

The missing object is often a contemporaneous decision record: an artefact showing the state presented to the control, the rule or authority applied, the outcome produced, and the content—if any—released beyond the control boundary.

ISO/IEC 42001 does not prevent such evidence. Its technology-neutral structure leaves the implementation choice to the organisation. Aurora-Lens addresses that choice at runtime.

2. Management-system evidence and decision-level evidence

Management-system evidence establishes that governance has been organised and operated. It may include:

- policies and assigned responsibilities;

- risk and impact assessments;
- treatment plans and operating procedures;
- training and competence records;
- monitoring results;
- internal audits;
- management reviews; and
- corrective-action records.

Decision-level evidence answers a different set of questions:

- What did the model propose?
- What governing state applied?
- Was the authority current, sufficient and unambiguous?
- Which control was evaluated?
- What outcome did the control produce?
- What was released, revised, contained or suppressed?
- Can the record be checked for alteration?

These forms of evidence are complementary. Organisational records explain the control environment. Runtime records show how that environment behaved in a particular event.

The distinction becomes material when a decision is disputed. A policy proves that an organisation intended a control to operate. A contemporaneous runtime record can help establish whether it did operate, which conditions it encountered, and what followed.

3. The runtime evidence gap

Several parts of an AIMS can benefit from decision-level evidence.

3.1 Risk assessment and treatment

Risk treatment translates an identified risk into a chosen response. The response may involve policy, training, review, technical controls or some combination of them.

Where treatment depends on inference-time behaviour, evidence of design is weaker than evidence of execution. A rule stating that stale authority must not support action is useful. A runtime record showing that stale authority was detected, demoted and prevented from authorising release is more specific.

Aurora-Lens can implement selected treatments as executable admission conditions. The organisation remains responsible for identifying the risk, choosing the treatment, validating the policy and evaluating residual risk.

3.2 AI system impact assessment

Impact assessment is prospective: it identifies and evaluates potential effects of an AI system on individuals, groups and society. Runtime evidence supplies a retrospective counterpart. It can show how a control associated with an identified impact behaved in an individual case.

The runtime record does not replace the assessment. It provides evidence with which the organisation can test whether its assumptions and treatments survive contact with operation.

3.3 Operational planning and control

Operational control requires more than elegant policy language. The relevant processes must be carried out under controlled conditions, and the organisation must retain sufficient information to support confidence in their operation.

For AI systems whose output can enter consequential workflows, a per-decision record can strengthen that confidence. It reduces dependence on sampled logs, human recollection and post-incident reconstruction.

3.4 Monitoring, internal audit and corrective action

Decision records can support monitoring by exposing patterns such as:

- repeated refusal or containment;
- recurring absence of authority;
- evidence becoming stale before use;
- unresolved referents or classifications;
- policy paths that produce excessive escalation; and
- mismatches between expected and actual governance outcomes.

The same records can support internal audit and corrective action. They provide event-level material for replay, root-cause analysis and policy revision. Their value depends on the completeness of the implementation, integrity of the recorded inputs, retention controls and the independence of verification.

4. What Aurora-Lens contributes

Aurora-Lens is a provider-agnostic governance layer positioned between an application and a model provider. The model produces a candidate response. Aurora-Lens determines whether that candidate may cross the commitment boundary under the governing state and policy.

Its relevant capabilities are these.

4.1 Admissibility before release

Candidate output is evaluated before it reaches the downstream application. The governing question is whether the proposed output has a sufficient basis and authority to become consequential in the present state.

This separates model generation from permission to commit. A fluent answer acquires no authority merely by having been generated.

4.2 Governed non-commitment

Aurora-Lens treats non-release as a valid governance result. Depending on the applicable policy, an evaluation may admit, require revision, contain, request clarification, escalate, suspend pending re-attestation, or stop.

This matters because many governance systems describe refusal procedurally while treating it operationally as an exception. A first-class non-commitment outcome makes the absence of permission explicit and auditable.

4.3 Persistent governing state

The architecture evaluates admissibility against persistent state rather than relying solely on context reconstructed for each request. Authority, identity, evidential status, unresolved ambiguity and relevant conditions can therefore remain governable across turns, sessions and workflows.

Persistence does not guarantee that the stored state is true. It preserves the state whose validity the system must evaluate.

4.4 Freshness and evidence demotion

Authority and evidence may be time-bound or condition-bound. Aurora-Lens can distinguish a record that remains informative from one that still has authority to support consequence.

When a basis becomes stale, the architecture can demote it from consequence-bearing authority to contextual information and route the decision to a defined outcome. The old record remains visible; its former authority does not silently survive.

4.5 Execution-time audit evidence

Each governance decision can generate a tamper-evident ledger entry recording the proposal, relevant governing state, evaluation path, outcome, and released or suppressed content.

Hash chaining and message authentication can make later alteration detectable. They do not prove that every input was true, every policy was sound or every event was captured. Those questions require separate controls. The ledger's narrower value is integrity and traceability for the events it records.

4.6 Reproducible policy evaluation

Where the relevant policy evaluation is deterministic, the same recorded state, policy version and input should reproduce the same governance result.

This property strengthens replay. It does not make the whole AI system deterministic: the upstream model may remain stochastic, external facts may change, and different captured state may properly produce a different outcome.

5. Contribution to selected ISO/IEC 42001 requirement areas

The following mapping is deliberately limited. It describes where Aurora-Lens may supply technical evidence within an AIMS. It does not assign conformity to the component.

Requirement area	Organisational responsibility	Possible Aurora-Lens contribution
AI risk assessment and treatment	Identify, analyse, evaluate and treat AI risk	Execute selected runtime treatments and record their outcomes
AI system impact assessment	Assess potential consequences and maintain an appropriate assessment process	Supply event-level evidence for testing how associated controls behaved
Documented information	Create, update, control and retain required information	Produce integrity-protected decision records, subject to the organisation's document controls
Operational planning and control	Plan, implement and control processes needed by the AIMS	Enforce admission conditions before model output reaches a consequential workflow
Monitoring, measurement, analysis and evaluation	Decide what to monitor, how, when and by whom	Supply decision telemetry and outcome data for selected runtime controls
Internal audit	Evaluate whether the AIMS conforms to organisational and standard requirements and is effectively implemented	Provide replayable event records as one source of audit evidence
Nonconformity and corrective action	Respond to nonconformity, determine causes and prevent recurrence where appropriate	Support reconstruction of the recorded decision path and comparison across events

The architecture contributes evidence. The organisation must still establish scope, policy, accountability, competence, objectives, risk criteria, impact-assessment methods, monitoring design, audit independence, management review and continual improvement.

6. Relevance to Annex A control families

Aurora-Lens has potential relevance to several Annex A control families, subject to the organisation's risk treatment decisions and statement of applicability.

Assessing impacts of AI systems

Runtime records can connect assessed impacts with observed control behaviour. They can help an organisation determine whether the operational evidence supports the assumptions made during assessment.

AI system life cycle

Aurora-Lens operates principally at deployment and use. It can preserve policy version, governing state and decision outcome as the system moves through operation and change. Development, verification and validation controls remain separate responsibilities.

Data for AI systems

The relevance is partial. Aurora-Lens can govern whether a stated evidential basis is admissible for a proposed

consequence and can demote stale evidence. It does not, by itself, establish the quality, representativeness, provenance or lawful use of training and operational data.

Information for interested parties

Decision records can support appropriately scoped explanations of what the runtime control did and why. Disclosure must still be governed by confidentiality, privacy, security, legal and usability requirements.

Use of AI systems

Aurora-Lens can make intervention points operational. Clarification, escalation, re-attestation and stop outcomes can be enforced as workflow states instead of expressed only as recommendations.

The architecture does not satisfy organisational controls concerning policy, internal organisation, resources or third-party relationships. It may generate evidence relevant to those controls, but evidence from a component is not performance of the organisational obligation.

7. What Aurora-Lens does not establish

Aurora-Lens does not certify an organisation to ISO/IEC 42001. Certification concerns the organisation's AIMS and requires assessment by a competent certification body where third-party certification is sought.

It does not determine the organisation's values, risk appetite or policy. It executes policy supplied to it. A defective policy can be enforced consistently and recorded beautifully. The ledger will then provide unusually good evidence of a bad decision.

It does not establish the truth of every external premise. Persistent state preserves identity and status across time; separate processes must establish and refresh the facts placed into that state.

It does not eliminate bias, unfairness or model error. These may become admission conditions where the organisation has defined testable controls, but the architecture does not make the underlying problems disappear.

It does not remove human responsibility. It can identify and enforce the points at which human judgment, escalation or renewed authority is required.

It does not make every part of an AI system deterministic. Its reproducibility claim applies to deterministic governance evaluation under the same captured inputs, state and policy version.

8. Evidentiary limits

Runtime evidence becomes probative only within a controlled evidentiary chain. A technically sophisticated ledger cannot rescue missing events, false inputs, compromised keys, inadequate retention, uncontrolled policy changes or an architecture that logs only favourable outcomes.

A credible implementation therefore needs, at minimum:

- defined event boundaries;
- reliable capture of relevant inputs and state;
- policy and configuration versioning;
- key management and access control;
- clock and ordering controls appropriate to the use case;
- retention and deletion rules;
- independent verification procedures;
- coverage testing for omitted paths;
- privacy and confidentiality safeguards; and
- documented treatment of failures in the governance layer itself.

Tamper evidence means that alteration can be detected under the stated integrity model. It does not mean that tampering is impossible, that the original event was correctly represented, or that the record is automatically admissible in legal proceedings.

This is where marketing language usually becomes drunk on its own adjectives. “Immutable,” “complete” and “proof” are conclusions that must be earned by the entire evidence chain.

9. The stronger implementation

ISO/IEC 42001 gives organisations a structure for governing AI responsibly. Its technology neutrality is necessary: a management-system standard must remain applicable across organisations, sectors and technical designs.

That neutrality leaves a practical choice.

An organisation can evidence selected controls primarily through policies, procedures, interviews, samples and retrospective investigation. It can also design those controls to produce contemporaneous technical evidence whenever model output approaches a consequential boundary.

The second approach does not replace the first. It gives the management system a memory of what its rules actually did.

Aurora-Lens is one architecture for creating that memory. It places an admissibility decision between generation and consequence, permits governed non-commitment, preserves the state against which authority is evaluated, and records the decision at the time it is made.

The resulting claim is precise:

Aurora-Lens can strengthen the implementation and auditability of selected ISO/IEC 42001 controls by producing decision-level evidence that a defined runtime governance process operated in a particular instance.

That claim is narrower than certification. It is also more useful than a badge. When consequences arrive, the decisive question is concrete: what was permitted, under whose authority, on which basis, and what evidence survives?

10. Related public records

- Margaret Stokes, *Aurora-Lens: A Provider-Agnostic Governance Proxy for Large Language Models*, 19 March 2026. <https://doi.org/10.5281/zenodo.19879302>
 - Margaret Stokes, *Epistemic Legitimacy as a Governance Layer for Large Language Models: Architecture and Implementation*, version 2, 11 April 2026. <https://doi.org/10.5281/zenodo.19504665>
 - Margaret Stokes, *When a Basis Goes Stale*, 2 July 2026. <https://doi.org/10.5281/zenodo.21123912>
 - Margaret Stokes, *Aurora-Lens: Proof and Precedence*, 16 July 2026. <https://doi.org/10.5281/zenodo.21397474>
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Publication note

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